

# Research Note on MathOverflow 413935

Exact relaxations, finite-temperature barriers, and a sharper proof frontier

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**Status and nonclaim.** This note does not prove that the limit in Question 1 exists, and it does not prove nonexistence. It contains independently checked partial theorems and exact finite computations. It must not be cited or submitted as a solution to the MathOverflow question.

## Abstract

For a symmetric zero-diagonal sign matrix  $A$ , put

$$M(A) = \max_{x \in \{-1,1\}^n} \left| \frac{1}{2} x^\top A x \right|, \quad F(n) = \min_A M(A).$$

The open question is whether  $F(n)/n^{3/2}$  converges. We first audit the previous bounds  $1/\pi \leq \liminf \leq \limsup \leq 1/2$ , the augmented cut-code identity, and one-vertex monotonicity; all survive audit. We then show that the proposed cube relaxation has the unique optimizer zero at zero temperature and at every fixed unscaled positive temperature, with a leading-order zero-temperature integrality gap. In contrast, an ellipsope relaxation has the exact thermodynamic limit  $1/2$ , while the defined two-covariance, edgewise-oriented Gaussian certificate has asymptotic lower-bound constant  $1/\pi$ . We identify the augmented cut code as a binary linear code whose dual consists of even-cardinality Eulerian subgraphs, and derive an exact signed-Eulerian MacWilliams formula. Ordinary graphons and limiting empirical spectra are proved to erase the  $n^{3/2}$  functional. Exact computer-assisted computation gives  $F(2), \dots, F(12) = 1, 3, 4, 4, 5, 9, 10, 12, 13, 17, 18$ . Exact cross-block computation shows that optimal blocks of orders 2 and 8 cannot occur together in an order-10 optimum. We also identify the one-vertex Bellman state as an exact energy-weighted projective covering radius and certify that the code of exact maximizers, and even the complete scalar partition-function curve, can be insufficient. Retaining the nonlinear arcsine term gives a trace-four bound; a parity theorem for its Gram defect yields  $F(20) \geq 30$  and  $F(21) \geq 32$ . The fully optimized one-parameter certificate nevertheless has exact asymptotic constant  $1/\pi$ . We derive an exact multivertex weighted Bellman identity, then prove that its weighted-entropy union-bound target has an unavoidable leading constant gap. The replacement is an exact switching orbit noncoverage problem on the projective rank-one group. A complete order-nine catalogue shows that even the energy histogram plus the two-point overlap law of exact maximizers does not determine extension, while the order-ten catalogue exhibits three exact finite-temperature optimizer phases. The annealed-normalized negative replica is exactly supermultiplicative; reverse hypercontractivity follows the required extensive parameter curve but in the wrong direction, isolating a power-saving transport conjecture whose finite defect remains visible through order nine. We also record the known regular Paley construction: square-order Paley conference matrices have Boolean eigenvectors attaining their spectral ceiling, a statement about those matrices rather than the minimax  $F$ . Finally, an explicit scalar countermodel shows that even uniform  $O(\sqrt{n})$  Bellman control does not force convergence. The sharp remaining target is a quenched minimax free-energy limit, or a genuinely dependence-sensitive composition theorem.

# 1 Problem, audited baseline, and scope

The question, asked by Paata Ivanisvili on MathOverflow in 2022 [1], is whether

$$\lim_{n \rightarrow \infty} \frac{F(n)}{n^{3/2}}$$

exists. Here  $A_{ii} = 0$ ,  $A_{ij} = A_{ji} = a_{ij} \in \{-1, 1\}$ , and

$$Q_A(x) = \sum_{i < j} a_{ij} x_i x_j = \frac{1}{2} x^\top A x.$$

The earlier attempt was rechecked from source rather than assumed. Its central conclusions are correct:

$$\frac{1}{\pi} \leq \liminf_{n \rightarrow \infty} \frac{F(n)}{n^{3/2}} \leq \limsup_{n \rightarrow \infty} \frac{F(n)}{n^{3/2}} \leq \frac{1}{2}, \quad (1)$$

$$F(n) = \binom{n}{2} - 2\rho(D_n), \quad F(n) \leq F(n+1) \leq F(n) + n. \quad (2)$$

The lower bound uses two Gaussian sign roundings and the arcsine law. The upper bound uses symmetric Paley conference matrices and principal submatrices. The code  $D_n$  is the cut-character set augmented by its negative. A claim-by-claim audit, including two attribution errors and a factor-of-two prose correction, is in `AUDIT.md`. Nothing in (1) or (2) implies convergence.

All new statements below distinguish a proved identity from a proposed limit theorem. Exact computations guide the search but are not used as asymptotic proofs.

## 2 The relaxed-coupling program collapses

Let  $m = \binom{n}{2}$ , let  $c_x = (x_i x_j)_{i < j}$ , and for arbitrary real couplings define

$$G_n(a) = \max_{x \in \{-1, 1\}^n} |\langle a, c_x \rangle|.$$

**Theorem 1** (Cube relaxation). *The function  $G_n$  is a norm on  $\mathbb{R}^m$ . Hence*

$$\min_{a \in [-1, 1]^m} G_n(a) = 0,$$

*and the unique minimizer is  $a = 0$ .*

*Proof.* Homogeneity and the triangle inequality follow from the maximum of absolute linear forms. If  $X$  is uniform on  $\{-1, 1\}^n$ , orthogonality of distinct degree-two Walsh characters gives

$$\mathbb{E} \langle a, c_X \rangle^2 = \sum_{i < j} a_{ij}^2.$$

Thus  $G_n(a) = 0$  forces  $a = 0$ . Since the origin belongs to the cube, the last assertion follows.  $\square$

In particular, any convex feasible set in the original coupling coordinates that contains every sign coupling contains their convex hull, the whole cube, and therefore contains this zero optimizer when the objective remains  $G_n$ . By the audited lower bound, its additive integrality gap is at least  $n\sqrt{n-1}/\pi$ , of the full target order.

The same collapse persists at positive temperature. Retain duplicated states  $(x, s) \in \{-1, 1\}^n \times \{-1, 1\}$  and put

$$L_{n,t}(a) = \frac{1}{t} \log \sum_{x,s} \exp(t \langle a, s c_x \rangle), \quad t > 0.$$

**Theorem 2** (Finite-temperature cube dual). *The function  $L_{n,t}$  is strictly convex and has the unique cube minimizer zero, with value*

$$\min_{a \in [-1,1]^m} L_{n,t}(a) = \frac{(n+1) \log 2}{t}.$$

*It has the exact dual representation*

$$\max_{p \in \Delta(\{-1,1\}^n \times \{-1,1\})} \left\{ \frac{H(p)}{t} - \|\mathbb{E}_p[sc_x]\|_1 \right\}.$$

*Proof.* The Hessian of  $L_{n,t}$  is  $t$  times the covariance matrix of  $sc_x$  under a full-support Gibbs distribution. If a linear functional of these vectors were constant, replacing  $s$  by  $-s$  would make the constant zero; Walsh orthogonality would then make the functional zero. The Hessian is therefore positive definite. Symmetry makes the gradient vanish at zero, where the value is  $t^{-1} \log 2^{n+1}$ .

The entropy variational formula for log-sum-exp gives

$$L_{n,t}(a) = \max_p \left\{ \langle a, \mathbb{E}_p[sc_x] \rangle + \frac{H(p)}{t} \right\}.$$

Both domains are compact and convex and the displayed function is convex in  $a$  and concave in  $p$ , so Sion's theorem applies. Minimization of a linear form over the cube is minus the  $\ell^1$ -norm of its coefficient.  $\square$

Thus the simplest probability-measure dual is exact but vacuous for the discrete problem: the uniform state measure cancels every first moment.

There is nevertheless a useful conditional rounding statement.

**Proposition 3** (Variance-sensitive sign rounding). *For  $a \in [-1,1]^m$ , set  $V(a) = \sum_e (1 - a_e^2)$ . There is a sign vector  $\sigma \in \{-1,1\}^m$  for which*

$$G_n(\sigma) \leq G_n(a) + \sqrt{2V(a)(n+1) \log 2} + \frac{4}{3}(n+1) \log 2.$$

*Proof.* Round the coordinates independently with  $\mathbb{E}\sigma_e = a_e$ . For fixed  $x$ , the centered error  $\sum_e (\sigma_e - a_e)c_x(e)$  has variance  $V(a)$  and summands bounded by two. The two-sided Bernstein bound with parameter  $u$  is at most  $2e^{-u}$  beyond  $\sqrt{2V(a)u} + 4u/3$ . There are  $2^{n-1}$  distinct cut characters. Taking  $u = (n+1) \log 2$  makes the union failure probability at most  $1/2$ . For a successful outcome, add this uniform error to  $G_n(a)$ .  $\square$

This gives  $o(n^{3/2})$  loss if  $V(a) = o(n^2)$ , but the cube optimizer has  $V(0) = m$ . The obvious spherical repairs also fail. On  $\|a\|_2 = \sqrt{m}$ , Parseval gives  $G_n(a) \geq \sqrt{m}$ , attained by a one-coordinate vector. Adding  $|a_e| \leq 1$  to that sphere forces every  $|a_e| = 1$ , recovering the original discrete problem exactly.

### 3 Temperature scaling and the cross-block wall

The functional proposed in the question is

$$\Phi_{n,\beta} = \min_A \frac{1}{\beta n^{3/2}} \log \sum_x e^{\beta |Q_A(x)|}.$$

For every fixed  $\beta > 0$ , the maximum-versus-log-sum-exp inequality gives

$$\frac{F(n)}{n^{3/2}} \leq \Phi_{n,\beta} \leq \frac{F(n)}{n^{3/2}} + \frac{\log 2}{\beta\sqrt{n}}. \quad (3)$$

Thus existence of  $\lim_n \Phi_{n,\beta}$  for even one fixed positive  $\beta$  is equivalent to the original limit. At this normalization, fixed temperature is already asymptotically zero temperature.

A genuinely extensive mean-field functional is

$$\Psi_{n,\beta} = \min_A \frac{1}{\beta n} \log \sum_{x,s} \exp \left( \frac{\beta s Q_A(x)}{\sqrt{n}} \right).$$

It satisfies

$$\frac{F(n)}{n^{3/2}} \leq \Psi_{n,\beta} \leq \frac{F(n)}{n^{3/2}} + \frac{(n+1) \log 2}{\beta n}. \quad (4)$$

Consequently, existence of  $\lim_n \Psi_{n,\beta}$  for arbitrarily large fixed  $\beta$  would settle the problem: the limsup-minus-liminf gap would be at most  $\log 2/\beta$  for every such  $\beta$ .

The direct interpolation does not have the needed accuracy. Define

$$h_n(t) = \min_A \log \sum_{x,s} e^{ts Q_A(x)}.$$

**Proposition 4** (Exact annealed block estimate). *For all  $n, k$  and  $t > 0$ ,*

$$h_{n+k}(t) \leq h_n(t) + h_k(t) - \log 2 + nk \log \cosh t.$$

*Proof.* For a within-block matrix  $A$ , write  $Z_A^s = \sum_x e^{ts Q_A(x)}$ . Replacing all coefficients of the second block by their negatives swaps its two one-sided partition sums. Of the two possible alignments, one satisfies

$$Z_A^+ Z_B^+ + Z_A^- Z_B^- \leq \frac{1}{2} (Z_A^+ + Z_A^-) (Z_B^+ + Z_B^-).$$

Choose every cross coefficient independently and uniformly. For fixed  $x, y, s$ , averaging the cross exponential gives  $(\cosh t)^{nk}$ . Therefore the expected augmented partition sum is at most the right side above times this factor, and some deterministic cross signing is no larger. Take logarithms and infima.  $\square$

At  $t = \beta/\sqrt{n+k}$  and  $n/(n+k) \rightarrow \alpha$ , the normalized cross cost tends to

$$\frac{\beta}{2} \alpha (1 - \alpha),$$

which is nonzero for fixed  $\beta$  and grows in the zero-temperature limit. The direct annealed cross block is therefore a leading term, not an error.

Finally, the absolute value cannot be silently removed. If every edge from vertex 1 is positive and every other edge negative, fix  $x_1 = 1$  and put  $T = \sum_{i>1} x_i$ . Then

$$Q_A(x) = \frac{n}{2} - \frac{(T-1)^2}{2}.$$

Hence  $\max_x Q_A(x) = \lfloor n/2 \rfloor$ , whereas  $\max_x |Q_A(x)| = \binom{n}{2}$ .

## 4 Covariance relaxations: two exact asymptotic limits

Let

$$\mathcal{E}_n = \{R \succeq 0 : \text{diag } R = \mathbf{1}\}$$

be the elliptope. The full arcsine expression is tautologically exact:

$$M(A) = \max_{R \in \mathcal{E}_n} \left| \frac{2}{\pi} \sum_{i < j} a_{ij} \arcsin R_{ij} \right|. \quad (5)$$

Indeed, Gaussian sign rounding makes every value on the right at most  $M(A)$ , while  $R = xx^\top$  attains  $Q_A(x)$ . The nonlinearity therefore preserves the original problem rather than relaxing it.

Linearizing gives a model with a complete thermodynamic limit. Put

$$S(A) = \max_{R \in \mathcal{E}_n} \left| \frac{1}{2} \text{tr}(AR) \right|, \quad S_n = \min_A S(A).$$

**Theorem 5** (Adversarial elliptope limit). *For every  $n \geq 2$ ,*

$$S_n \geq \frac{1}{2} n \sqrt{n-1},$$

and

$$\boxed{\lim_{n \rightarrow \infty} \frac{S_n}{n^{3/2}} = \frac{1}{2}}.$$

*At each symmetric conference order, equality holds in the finite bound.*

*Proof.* Fix any sign matrix  $A$ , set  $q = n - 1$ , and define

$$R^\pm = \frac{1}{2} \left( I \pm \frac{A}{\sqrt{q}} \right)^2.$$

They are positive semidefinite. Since every row of  $A$  has squared norm  $q$ , their diagonals equal one. Moreover

$$R^+ - R^- = \frac{2A}{\sqrt{q}}.$$

The corresponding two linear SDP values differ by

$$\frac{1}{2} \text{tr}(A(R^+ - R^-)) = n\sqrt{q}.$$

At least one has absolute value at least half this amount.

For the reverse asymptotic inequality, if  $R \in \mathcal{E}_n$ , then

$$|\text{tr}(AR)| \leq \|A\|_{\text{op}} \text{tr } R = n\|A\|_{\text{op}}.$$

A symmetric conference matrix has operator norm  $\sqrt{n-1}$ , so it attains equality. For general  $n$ , the prime number theorem in arithmetic progressions supplies a prime  $q \equiv 1 \pmod{4}$  with  $q+1 \geq n$  and  $q+1 = n(1+o(1))$  [12]. Take a principal  $n$ -by- $n$  submatrix of the symmetric Paley conference matrix of order  $N = q+1$ . Compression does not increase its operator norm  $\sqrt{N-1}$ , giving  $S_n \leq n\sqrt{N-1}/2 = (1/2 + o(1))n^{3/2}$ .  $\square$

Rank-one  $R = xx^\top$  show  $S(A) \geq M(A)$ . The missing transfer is an asymptotically vanishing integrality gap, which is not supplied by standard Grothendieck-type constant-factor rounding [5, 6]. The gap is already strict at order six: the exact computation below gives  $F(6) = 5$ , whereas Theorem 5 and a conference matrix give  $S_6 = 3\sqrt{5}$ .

There is also a sharp barrier for the linear Gaussian-sign method. Let  $\Lambda(A)$  be the maximum of

$$\sum_{i < j} a_{ij}(R_{ij}^+ - R_{ij}^-)$$

over  $R^\pm \in \mathcal{E}_n$ , subject to the edgewise orientations  $a_{ij}(R_{ij}^+ - R_{ij}^-) \geq 0$ .

**Theorem 6** (Oriented covariance barrier). *For every sign matrix  $A$ ,*

$$M(A) \geq \frac{\Lambda(A)}{\pi}, \quad \Lambda(A) \geq n\sqrt{n-1}.$$

Moreover

$$\boxed{\lim_{n \rightarrow \infty} \frac{\min_A \Lambda(A)}{n^{3/2}} = 1.}$$

*Proof.* Gaussian sign rounding of  $R^+$  and  $R^-$ , followed by the arcsine law, gives two expected energies. Since  $\arcsin'(r) \geq 1$  and each edge difference has the prescribed orientation, their difference is at least  $2\Lambda(A)/\pi$ . Both expectations lie in  $[-M(A), M(A)]$ , proving the first inequality.

The square covariances used in Theorem 5 satisfy the orientation constraints and have objective  $n\sqrt{n-1}$ . Conversely,

$$\Lambda(A) = \frac{1}{2} \operatorname{tr}(A(R^+ - R^-)) \leq n\|A\|_{\text{op}}.$$

Conference matrices attain equality. The same prime-number-theorem and Paley principal-submatrix construction used in Theorem 5 gives the matching asymptotic upper bound at every order.  $\square$

Thus the optimized two-covariance, edgewise-oriented linear certificate  $\Lambda$  defined above has adversarial asymptotic lower-bound constant exactly  $1/\pi$ . Improving it requires information beyond this specific linearization; the theorem does not exclude more general covariance certificates.

## 5 The augmented cut code and its exact dual

Encode signs as bits via  $(-1)^b$ . Let  $j \in \mathbb{F}_2^m$  be the all-one edge word and

$$(\delta u)_{ij} = u_i + u_j.$$

Then the augmented code in (2) is

$$D_n = \{\delta u + sj : u \in \mathbb{F}_2^n, s \in \mathbb{F}_2\}.$$

**Theorem 7** (Linear code and dual). *For  $n \geq 3$ ,  $D_n$  is a binary  $[m, n]$  linear code and*

$$D_n^\perp = \{H \subseteq E(K_n) : \deg_H(v) \equiv 0 \pmod{2} \text{ for all } v, |H| \equiv 0 \pmod{2}\}.$$

*For  $n \geq 4$ , the dual is spanned by 4-cycles, has dimension  $m - n$ , and minimum distance four.*

*Proof.* The cut space has dimension  $n - 1$ . Its words have even parity on every triangle, whereas  $j$  has odd triangle parity, so adjoining  $j$  raises the dimension by one. A graph is orthogonal to every cut exactly when every vertex has even degree, and it is orthogonal to  $j$  exactly when its number of edges is even.

The cycle space is generated by triangles through one fixed vertex. An even-weight cycle-space word is a sum of an even number of these basis triangles. The sum of two sharing a nonfixed vertex is a 4-cycle. Two that share only the fixed vertex can be paired through an intermediate basis triangle and expressed as two 4-cycles. This proves generation. A nonzero even-cardinality Eulerian graph has at least four edges, and a 4-cycle exists.  $\square$

Counting cuts by the size of one shore gives

$$W_{D_n}(z) = \frac{1}{2} \sum_{k=0}^n \binom{n}{k} \left( z^{k(n-k)} + z^{m-k(n-k)} \right). \quad (6)$$

Let  $a_e = (-1)^{y_e}$  and define the exact augmented partition function

$$Z_y(\beta) = \sum_{d \in D_n} \exp(\beta[m - 2 \text{wt}(y + d)]).$$

Its ground state is  $M(A)$ .

**Theorem 8** (Signed-Eulerian MacWilliams formula). *For  $n \geq 3$ ,*

$$Z_y(\beta) = 2^n (\cosh \beta)^m P_y(\tanh \beta),$$

where

$$P_y(t) = \sum_{H \in D_n^\perp} (-1)^{|H \cap y|} t^{|H|}.$$

For  $0 \leq t < 1$ ,  $P_y(t) > 0$ , and uniformly in  $y$ ,

$$M(A) \leq \frac{1}{\beta} \log Z_y(\beta) \leq M(A) + \frac{n \log 2}{\beta}.$$

*Proof.* The coset MacWilliams identity is

$$W_{y+D_n}(z) = 2^{n-m} (1+z)^m \sum_{H \in D_n^\perp} (-1)^{|H \cap y|} \left( \frac{1-z}{1+z} \right)^{|H|}.$$

Multiply by  $e^{\beta m}$  and substitute  $z = e^{-2\beta}$ . This gives the formula because  $e^\beta(1 + e^{-2\beta}) = 2 \cosh \beta$  and the final ratio is  $\tanh \beta$ . Positivity follows because the left side is a positive partition sum. The last display compares its largest summand with a sum of  $|D_n| = 2^n$  summands.  $\square$

For fixed  $\beta > 0$ , the minimized free energy in Theorem 8, divided by  $n^{3/2}$ , differs from  $F(n)/n^{3/2}$  by  $O(n^{-1/2})$ . Thus the exact unresolved term is

$$\min_y \log P_y(\tanh \beta).$$

It must cancel the explicit  $m \log \cosh \beta$  term to  $n^{3/2}$  accuracy. Averaging  $P_y(t)$  over all  $y$  gives one and yields only

$$F(n) \leq n \sqrt{(n-1) \log 2},$$

the usual sphere/union-bound constant, weaker than  $1/2$ .

Puncturing also stops at the old scale. For  $n \geq 4$ , deleting the final star maps  $D_n$  onto  $D_{n-1}$ , and the two lifts of each word have complementary star coordinates. Therefore

$$\rho(D_{n-1}) \leq \rho(D_n) \leq \rho(D_{n-1}) + \left\lfloor \frac{n-1}{2} \right\rfloor,$$

equivalently

$$F(n-1) + ((n-1) \bmod 2) \leq F(n) \leq F(n-1) + n-1.$$

The  $n = 3$  instance of these inequalities is checked directly; the fiber there has four rather than two words. Across a two-block partition, the cross lifts are complementary rank-one sign matrices. On the dual side, the  $\binom{n}{2} \binom{k}{2}$  alternating cross-block 4-cycles already contribute macroscopically at mean-field scaling. Product bounds that take absolute values discard the cancellation needed here.

## 6 High-temperature data do not truncate uniformly

Let  $X$  be uniform on  $\{-1, 1\}^n$ , let  $Y = Q_A(X)$ , and use the symmetrized partition function  $\mathbb{E} \cosh(tY)$ , equivalently the augmented state  $sY$ . Direct Walsh counting gives

$$\mathbb{E} Y^2 = m, \quad \mathbb{E} Y^4 = 3m^2 - 2m + 24 \sum_{C_4} \prod_{e \in C_4} a_e, \quad (7)$$

where the sum is over undirected 4-cycles. Indeed, a monomial in the fourth power survives exactly when every vertex has even incidence. Besides one edge four times and two edges twice, the only case is a simple 4-cycle, with multinomial factor 24.

Equivalently,

$$\kappa_4(Y) = 3 \operatorname{tr}(A^4) - 2n(n-1)(3n-4), \quad \operatorname{tr}(A^4) = n(n-1)^2 + \sum_{i \neq j} (A_{ij}^2)^2. \quad (8)$$

The second identity is  $\operatorname{tr}(A^4) = \sum_{ij} (A_{ij}^2)^2$ ; the first follows by separating the closed walks with four distinct vertices in (7).

The expansion

$$\log \mathbb{E} \cosh(tY) = \frac{mt^2}{2} + \frac{\kappa_4(Y)t^4}{24} + O(t^6)$$

shows that, for each fixed  $n$ , all sufficiently high-temperature minimizers of this symmetrized partition function minimize  $\operatorname{tr}(A^4)$ . The symmetrization matters: the one-sided partition can have a signing-dependent cubic triangle term. Spectral Cauchy–Schwarz gives

$$\operatorname{tr}(A^4) \geq \frac{\operatorname{tr}(A^2)^2}{n} = n(n-1)^2.$$

For a Paley principal submatrix of order  $n$  inside a conference matrix of order  $N = n(1 + o(1))$ ,

$$\operatorname{tr}(A^4) \leq \|A\|_{\text{op}}^2 \operatorname{tr}(A^2) \leq (N-1)n(n-1).$$

This gives the reverse asymptotic inequality, so

$$\lim_n \frac{\min_A \operatorname{tr}(A^4)}{n^3} = 1, \quad \lim_n \frac{\min_A \kappa_4(Y)}{n^3} = -3.$$



This fourth-order limit does not determine the next term. Exhausting all root-normalized order-seven signings gives

$$\min_A \text{tr}(A^4) = 342,$$

while the trace-minimizing signings have three distinct exact sixth moments,

$$50,781, \quad 53,661, \quad 59,421.$$

Thus fourth-order data alone are insufficient. No uniform remainder theorem connecting this high-temperature expansion to the required ground-state regime was obtained.

## 7 Ordinary graphons and empirical spectra lose the functional

For a signed matrix  $A$ , let  $W_A$  be its step graphon and normalize

$$\|W_A\|_{\square} = \frac{1}{n^2} \max_{S, T \subseteq [n]} \left| \sum_{i \in S, j \in T} a_{ij} \right|.$$

**Theorem 9** (Cut-limit obstruction). *If  $M(A_n) = O(n^{3/2})$ , then  $W_{A_n} \rightarrow 0$  in cut norm. There are two deterministic sequences satisfying this hypothesis and having the same zero graphon limit, while their normalized Boolean maxima are separated by at least  $3/2$ .*

*Proof.* For  $x, y \in \{-1, 1\}^n$ , split the coordinates according to whether  $x_i = y_i$ . Writing  $z^+, z^- \in \{-1, 0, 1\}^n$  for the corresponding restrictions of  $y$ , symmetry of  $A$  cancels cross terms and gives

$$x^T A y = 2Q_A(z^+) - 2Q_A(z^-).$$

A multilinear quadratic form attains its maximum absolute value on  $[-1, 1]^n$  at a Boolean vertex, so every sign bilinear form is bounded by  $4M(A)$ . Expressing two indicator vectors as affine combinations of sign vectors then gives

$$\|W_A\|_{\square} \leq \frac{4M(A)}{n^2}.$$

Take an infinite sequence of Paley conference matrices  $C_N$ . Their normalized Boolean maxima have limsup at most  $1/2$ , and their graphons tend to zero. Choose  $k = \lfloor 2N^{3/4} \rfloor$  vertices and change every coefficient inside that set to  $+1$ , obtaining  $B_N$ . At most  $\binom{k}{2} = O(N^{3/2})$  edges change, so  $M(B_N) = O(N^{3/2})$  and the graphon perturbation tends to zero in  $L^1$ . Fix the planted spins to  $+1$  and average all outside spins independently. Every cross and outside term has mean zero, while the planted block contributes  $\binom{k}{2}$ . Some outside assignment therefore has at least that energy, and

$$\liminf_N \frac{M(B_N)}{N^{3/2}} \geq 2.$$

The separation from the conference limsup is at least  $3/2$ . □

The usual dense graph topology therefore collapses every target-scale near-optimizer to the same point. This is not repaired by keeping only a limiting empirical spectrum. At dense scaling, every empirical law of  $A/n$  tends to  $\delta_0$ , since its average squared eigenvalue is  $(n-1)/n^2$ . At fluctuation scaling there is a stronger obstruction.

**Proposition 10** (Empirical-spectral obstruction). *For every fixed  $d > 1/2$ , there are conference matrices  $C_N$  and sign matrices  $B_N$  such that*

$$W_2(\mu_{B_N/\sqrt{N}}, \mu_{C_N/\sqrt{N}}) \longrightarrow 0,$$

but

$$\liminf_N \frac{M(B_N)}{N^{3/2}} \geq d, \quad \limsup_N \frac{M(C_N)}{N^{3/2}} \leq \frac{1}{2}.$$

*Proof.* Let  $q_N = Q_{C_N}(\mathbf{1})$ . The spectral bound gives  $|q_N| \leq N\sqrt{N-1}/2$ . Flip negative coefficients to positive until the all-one energy is at least  $dN^{3/2}$ . This needs  $O(N^{3/2})$  edges, and there are  $\Theta(N^2)$  available. Therefore  $\|B_N - C_N\|_F^2 = O(N^{3/2})$ . Hoffman–Wielandt implies

$$W_2^2(\mu_{B_N/\sqrt{N}}, \mu_{C_N/\sqrt{N}}) \leq \frac{\|B_N - C_N\|_F^2}{N^2} = O(N^{-1/2}).$$

The energy inequalities follow from the construction and the conference spectral bound.  $\square$

This excludes dense graphons and limiting empirical spectral distributions, not spectral edges, eigenvectors, or a richer Gaussian-process limit.

For comparison, iid signings are rigorously worse than the conference upper constant.

**Proposition 11** (Greedy energy of an iid signing). *On one infinite iid Rademacher edge array  $A_n$ , almost surely*

$$\liminf_n \frac{M(A_n)}{n^{3/2}} \geq \frac{2}{3} \sqrt{\frac{2}{\pi}} = 0.531923 \dots > \frac{1}{2}.$$

*Proof.* Choose spins sequentially by

$$x_1 = 1, \quad x_j = \operatorname{sgn} \sum_{i < j} a_{ij} x_i.$$

The resulting energy is

$$Q_{A_n}(x) = \sum_{j=2}^n \left| \sum_{i < j} a_{ij} x_i \right| = \sum_{k=1}^{n-1} Y_k.$$

Before column  $j$  is exposed, the earlier spins are measurable with respect to earlier columns. Multiplying the fresh iid column signs by those predictable spins preserves their iid Rademacher law. Hence the  $Y_k$  are independent and  $Y_k$  has the law of  $|\varepsilon_1 + \dots + \varepsilon_k|$ .

The central limit theorem and uniform integrability give  $\mathbb{E}Y_k/\sqrt{k} \rightarrow \sqrt{2/\pi}$ . A Riemann-sum argument therefore yields

$$\frac{1}{n^{3/2}} \sum_{k < n} \mathbb{E}Y_k \longrightarrow \sqrt{\frac{2}{\pi}} \int_0^1 \sqrt{t} dt = \frac{2}{3} \sqrt{\frac{2}{\pi}}.$$

Changing one input sign changes  $Y_k$  by at most two, so its centered moment generating function is bounded by  $\exp(\lambda^2 k/2)$ . Independence gives an exponential tail  $2e^{-c_\varepsilon n}$  for deviations of  $\sum_{k < n} Y_k$  by  $\varepsilon n^{3/2}$ . This is summable; Borel–Cantelli completes the proof.  $\square$

This is not a lower bound on  $F(n)$ ; it shows that typical iid signings are not asymptotically optimal.

## 8 Exact computation through order twelve

Switching makes every edge incident with vertex zero positive. The remaining negative coefficients form a graph on  $n - 1$  vertices, and vertex permutations reduce the search to one unlabeled graph in each isomorphism class. For a residual negative-edge graph  $G$  and a negative-spin set  $S$  not containing the root, the exact energy is

$$\binom{n}{2} - 2|E(G)| - 2|S|(n - |S|) + 4|\delta_G(S)|.$$

Exhausting nauty’s **geng** streams gives Table 1.

Table 1: Exact computational values.

| $n$            | 2     | 3     | 4     | 5     | 6     | 7     | 8     | 9     | 10    | 11    | 12    |
|----------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| $F(n)$         | 1     | 3     | 4     | 4     | 5     | 9     | 10    | 12    | 13    | 17    | 18    |
| $F(n)/n^{3/2}$ | .3536 | .5774 | .5000 | .3578 | .3402 | .4860 | .4419 | .4444 | .4111 | .4660 | .4330 |

Completeness is checked against the unlabeled-graph counts and hashes of the full graph streams. The reduced formula is checked by direct evaluation; all labeled signings are independently enumerated through  $n = 6$ ; optional NetworkX and Z3 routes use different decoders and optimization mechanisms. Z3 independently gives UNSAT/SAT pairs certifying the displayed values for  $n = 7, 8, 9$ . At  $n = 10$ , the lower certificate is the exhaustive nauty search. At order 11, the full stream of 12,005,168 unlabeled residual graphs is hashed and exactly filtered to the classes that can have maximum at most 15. A reduced stream with  $20 \leq |E(G)| \leq 22$  and maximum degree at most six contains 2,153,606 records and reproduces the zero-survivor result with the same evaluator. The two stream hashes and the exact cut computation are recorded in `verification/research_order11_certify.py`. Deterministic lower-scan samples at edge counts 20, 21, and 22 are also recomputed by an adjacency formula. The order-11 graph6 witness `ICRbczQMo` has maximum 17. Puncturing and energy parity then give  $F(12) \geq 18$ , while the witness `JWUuDOR\K{?}` has maximum 18. Completeness at orders 10 and 11 trusts nauty; no solver timeout is used in either result.

The Paley conference matrix of order ten over  $\mathbb{F}_9$  has exact Boolean maximum 15, strictly worse than  $F(10) = 13$ . The two optimal switching classes at order ten have different characteristic polynomials. Thus neither conference structure nor a unique spectrum describes all finite optimizers. These finite facts do not imply a limiting constant.

## 9 Finite non-heredity and an exact weighted covering state

For fixed signings  $A$  and  $B$ , define

$$J(A, B) = \min_{C \in \{-1, 1\}^{s \times k}} M \begin{pmatrix} A & C \\ C^\top & B \end{pmatrix}, \quad E(B) = J([0], B).$$

Complete cross-block enumeration over the optimal switching-permutation classes through total order ten gives, in particular,

$$\min_{M(A)=F(2), M(B)=F(8)} J(A, B) = 15 > F(10) = 13. \quad (9)$$

Hence no optimal order-10 signing contains an optimal order-8 principal submatrix. The smallest possible internal sacrifice repairs the obstruction: among the 1,044 root-normalized order-8 representatives, internal maximum 12 is necessary and sufficient for a  $2 + 8$  completion of maximum 13.

The same enumeration gives one-vertex extension distributions

| $n$ | distribution of $E(B)$ over optimal classes |
|-----|---------------------------------------------|
| 7   | $10^{\times 4}, 12^{\times 2}$              |
| 8   | $12^{\times 2}$                             |
| 9   | $13^{\times 4}, 15^{\times 11}$             |
| 10  | $17^{\times 1}, 19^{\times 1}$ .            |

Thus the scalar  $M(B)$  does not determine future composition cost.

There is, however, an exact coding-theoretic state for one-vertex extension. Let

$$\mathcal{P}_n = \{-1, 1\}^n / \{x \sim -x\}, \quad d_{\pm}([u], [v]) = \min\{d_H(u, v), d_H(u, -v)\}.$$

For an order- $n$  signing  $B$ , put  $M = M(B)$  and

$$w_B([x]) = \frac{M - |Q_B(x)|}{2}, \quad \rho_w(B) = \max_{[b] \in \mathcal{P}_n} \min_{[x] \in \mathcal{P}_n} (d_{\pm}([b], [x]) + w_B([x])).$$

The weights are nonnegative integers because every quadratic energy has the parity of  $\binom{n}{2}$ .

**Theorem 12** (Weighted covering-radius Bellman identity). *For every order- $n$  signing  $B$ ,*

$$E(B) = M(B) + n - 2\rho_w(B).$$

*Equivalently, if*

$$\delta_w(B) = \left\lfloor \frac{n}{2} \right\rfloor - \rho_w(B),$$

*then*

$$F(n+1) = \min_B (M(B) + (n \bmod 2) + 2\delta_w(B)),$$

*where the minimum is over all order- $n$  signings.*

*Proof.* Let  $b \in \{-1, 1\}^n$  be the incident signs and let  $t$  be the new spin. Pairing  $t = 1$  and  $t = -1$  gives, for every old spin vector  $x$ ,

$$\max_{t \in \{-1, 1\}} |Q_B(x) + t b \cdot x| = |Q_B(x)| + |b \cdot x|.$$

Since  $|b \cdot x| = n - 2d_{\pm}([b], [x])$ , the maximum over  $x$  for fixed  $b$  is

$$M + n - 2 \min_{[x] \in \mathcal{P}_n} (w_B([x]) + d_{\pm}([b], [x])).$$

Minimizing over  $[b]$  proves the first identity. The equality  $n - 2\lfloor n/2 \rfloor = n \bmod 2$  gives the deficit form. Finally, minimizing over all  $B$  and  $b$  enumerates every order- $(n+1)$  signing.  $\square$

For every center  $[b]$ , some exact maximizer has projective distance at most  $\lfloor n/2 \rfloor$ . A configuration of weight greater than this can never attain the inner minimum. Consequently  $E(B)$  depends only on the exact near-ground window

$$|Q_B(x)| \geq M(B) - 2 \left\lfloor \frac{n}{2} \right\rfloor.$$

The Bellman state has energy width at most  $n$ ; deeper configurations are rigorously irrelevant to one-vertex extension.

Let  $X_B = \{[x] : |Q_B(x)| = M(B)\}$  and let  $\rho_{\text{ext}}(B)$  be its ordinary covering radius in  $\mathcal{P}_n$ . The same proof, restricted to the maximizing configurations, gives the exact lower obstruction

$$E(B) \geq M(B) + n - 2\rho_{\text{ext}}(B), \quad \rho_w(B) \leq \rho_{\text{ext}}(B).$$

The exhaustive verifier independently recomputes both radii and the full extension values. At order nine, the four extendible classes have  $(\rho_{\text{ext}}, \rho_w) = (4, 4)$ , while the eleven failing classes have  $(3, 3)$ . The order-10 classes have radii  $(3, 3)$  and  $(2, 2)$ , exactly explaining extension values 17 and 19. Exact maximizers are not a complete state: all six order-7 classes have  $\rho_{\text{ext}} = 3$ , but the  $|Q_B| = 7$  layer lowers  $\rho_w$  to 2 in precisely the two classes with  $E(B) = 12$ . This is a finite counterexample to any cavity rule retaining only the ground-state code. The full energy-weighted projective landscape is exact for one step.

The scalar partition function is also insufficient. The two optimal order-nine graph6 records G?qmaw and GCpbaw have the same complete projective absolute-energy histogram

$$\#\{|Q_B| = 0, 4, 8, 12\} = (60, 111, 60, 25),$$

and therefore have identical values of  $\sum_x e^{t|Q_B(x)|}$  and  $\sum_{x,s} e^{tsQ_B(x)}$  for every real  $t$ . Nevertheless their extremizer covering radii are 4 and 3, and direct enumeration gives extension values 13 and 15. Thus even the entire scalar free-energy curve loses the geometry required by a Bellman step. This does not refute a nonlocal minimax interpolation with a richer order parameter.

The verifier also exhausts all root-normalized residual graphs through order eight. If  $\mathcal{B}_n$  is the coordinatewise Pareto-minimal set of pairs  $(M(B), \delta_w(B))$ , then

| $n$ | $\mathcal{B}_n$              |
|-----|------------------------------|
| 2   | $\{(1, 1)\}$                 |
| 3   | $\{(3, 0)\}$                 |
| 4   | $\{(4, 0)\}$                 |
| 5   | $\{(4, 0)\}$                 |
| 6   | $\{(5, 2), (7, 1), (9, 0)\}$ |
| 7   | $\{(9, 0)\}$                 |
| 8   | $\{(10, 1), (12, 0)\}$ .     |

Every point at order six gives  $F(7) = 9$ , and both points at order eight give  $F(9) = 12$ . Thus two units of additional internal energy can be exactly offset by one unit of covering-deficit improvement. This is finite evidence for a near-optimal Pareto family, not an asymptotic conclusion.

## 9.1 The nonlinear square-covariance gain

The earlier Gaussian proof used only the inequality  $\arcsin'(u) \geq 1$ . Retaining the discarded nonlinear term gives a stability theorem.

**Theorem 13** (Nonlinear Gaussian-sign bound). *For  $s \geq 0$ , put*

$$D_s = 1 + s^2(n-1), \quad d_s = \frac{2s}{D_s}, \quad z_{ij,s} = \frac{s^2|(A^2)_{ij}|}{D_s}.$$

*Then every order- $n$  signing satisfies*

$$M(A) \geq \frac{1}{\pi} \sum_{i < j} [\arcsin(z_{ij,s} + d_s) - \arcsin(z_{ij,s} - d_s)]. \quad (25)$$

Consequently

$$M(A) \geq \frac{n(n-1)}{\pi} \arcsin \frac{1}{\sqrt{n-1}}, \quad (26)$$

and, for  $n \geq 3$ ,

$$M(A) \geq \frac{n(n-1)}{\pi} \arcsin \frac{1}{\sqrt{n-1}} + \frac{\text{tr}(A^4) - n(n-1)^2}{8\pi(n-1)(n-2)^{3/2}}. \quad (27)$$

*Proof.* The two matrices

$$R_s^\pm = \frac{(I \pm sA)^2}{D_s}$$

are correlation matrices. With  $u_{ij} = s^2 a_{ij}(A^2)_{ij}/D_s$ , the corresponding Gaussian sign roundings and the arcsine law give

$$\mathbb{E}Q_A(X^+) - \mathbb{E}Q_A(X^-) = \frac{2}{\pi} \sum_{i < j} [\arcsin(u_{ij} + d_s) - \arcsin(u_{ij} - d_s)].$$

The bracket is even in  $u_{ij}$ , and each expectation belongs to  $[-M(A), M(A)]$ , proving (25).

For  $f_d(z) = \arcsin(z+d) - \arcsin(z-d)$  and  $z \geq 0$ ,

$$f_d''(z) = h(z+d) - h(z-d), \quad h(t) = \frac{t}{(1-t^2)^{3/2}}.$$

Thus  $f_d$  is even and convex. At  $s = 1/\sqrt{n-1}$  one has  $D_s = 2$  and  $d_s = 1/\sqrt{n-1}$ , so  $f_d(z) \geq f_d(0) = 2 \arcsin d$ , which proves (26).

Write  $q = n-1$ . Since

$$h'(t) = \frac{1+2t^2}{(1-t^2)^{5/2}}$$

is even and increasing in  $|t|$ ,  $f_d''(z)$  is increasing for  $z \geq 0$ . For  $d = 1/\sqrt{q}$ ,

$$f_d(z) \geq 2 \arcsin d + \frac{q}{(q-1)^{3/2}} z^2.$$

Finally  $z_{ij} = |(A^2)_{ij}|/(2q)$  and

$$\sum_{i < j} z_{ij}^2 = \frac{\text{tr}(A^4) - nq^2}{8q^2},$$

which proves (27). □

The excess is exactly

$$\text{tr}(A^4) - n(n-1)^2 = \|A^2 - (n-1)I\|_F^2.$$

Hence a sequence attaining the asymptotic lower constant  $1/\pi$  must obey  $\|A_n^2 - (n-1)I\|_F = o(n^2)$ . This is vanishing normalized Gram defect, not a conference-rigidity conclusion: a uniform random signing already satisfies this normalization in probability. The universal improvement in (26) is only  $\Theta(\sqrt{n})$ , so it does not change the leading constant. It does imply the new exact consequence

$$\boxed{F(21) \geq 32}.$$

Indeed (26) is greater than 30 and all order-21 energies are even. For an exact strictness certificate, use  $\pi/14 < 11/49$  and  $\sin t < t - t^3/6 + t^5/120$ ; rational squaring shows that the latter polynomial at  $11/49$  has square less than  $1/20$ . Thus  $\arcsin(1/\sqrt{20}) > \pi/14$ .

## 9.2 The multivertex Bellman state and its entropy bound

For internal signings  $B, D$  of orders  $n, k$ , define

$$J(B, D) = \min_C M \begin{pmatrix} B & C \\ C^\top & D \end{pmatrix}, \quad L(B, D) = \max_{x, y} |Q_B(x) + Q_D(y)|.$$

In the projective Hamming cube  $\mathcal{P}_{nk} = \{\pm 1\}^{n \times k} / \{C \sim -C\}$ , let

$$\mathcal{R}_{n, k} = \{[xy^\top] : x \in \{-1, 1\}^n, y \in \{-1, 1\}^k\}$$

and give  $R = [xy^\top]$  the integer weight

$$w_{B, D}(R) = \frac{L(B, D) - |Q_B(x) + Q_D(y)|}{2}.$$

Put

$$\rho_w^\square(B, D) = \max_{[C] \in \mathcal{P}_{nk}} \min_{R \in \mathcal{R}_{n, k}} [d_\pm([C], R) + w_{B, D}(R)].$$

**Theorem 14** (Multivertex weighted Bellman identity). *One has*

$$J(B, D) = L(B, D) + nk - 2\rho_w^\square(B, D). \quad (28)$$

Thus, with

$$\delta_w^\square = \left\lfloor \frac{nk}{2} \right\rfloor - \rho_w^\square,$$

$$F(n + k) = \min_{B, D} [L(B, D) + (nk \bmod 2) + 2\delta_w^\square(B, D)]. \quad (29)$$

*Proof.* Pairing  $y$  with  $-y$  gives

$$\max_{\eta = \pm 1} |Q_B(x) + Q_D(y) + \eta x^\top C y| = |Q_B(x) + Q_D(y)| + |x^\top C y|.$$

Moreover  $|x^\top C y| = nk - 2d_\pm([C], [xy^\top])$ . Substitute  $|Q_B + Q_D| = L - 2w$ , maximize over the rank-one code, and minimize over  $[C]$  to obtain (28). A weight-zero rank-one word is within projective distance  $\lfloor nk/2 \rfloor$  of every center, so the deficit is nonnegative. Minimizing over the two internal blocks proves (29).  $\square$

Only the internal window

$$|Q_B(x) + Q_D(y)| \geq L(B, D) - 2 \left\lfloor \frac{nk}{2} \right\rfloor$$

can affect this radius. Define its soft effective size by

$$\Xi(B, D) = \log \sum_{R \in \mathcal{R}_{n, k}} \exp \left( -\frac{2w_{B, D}(R)^2}{nk} \right).$$

**Proposition 15** (Weighted entropy upper bound).

$$J(B, D) \leq L(B, D) + \sqrt{2nk(\Xi(B, D) + \log 4)}. \quad (30)$$

*Proof.* Choose  $C$  uniformly at random. Hoeffding's inequality and a union bound give

$$\Pr\{\exists R : |\langle C, R \rangle| > T + 2w(R)\} \leq 2 \exp\left(-\frac{T^2}{2nk} + \Xi(B, D)\right).$$

For  $T = \sqrt{2nk(\Xi + \log 4)}$  this is at most  $1/2$ . Hence some  $C$  satisfies every displayed inequality, and  $L - 2w(R) + |\langle C, R \rangle| \leq L + T$  for every  $R$ . Theorem 14 finishes the proof.  $\square$

The generic bound  $\Xi \leq (n + k - 2) \log 2$  reproduces the known leading  $\sqrt{nk(n + k)}$  block cost. The initially proposed replacement was the following constant-matching profile law:

$$L(B, D) + \sqrt{2nk(\Xi(B, D) + \log 4)} \leq (F(n)^{2/3} + F(k)^{2/3})^{3/2} + O((n + k)^{3/2-\varepsilon}). \quad (31)$$

This would imply power-saving near-subadditivity of  $F^{2/3}$ , which the standard blocking argument turns into convergence. The next proposition shows that this initially proposed target is actually impossible; it is retained here as an audited failed route.

There is also density-one control of the exact one-vertex state. Choose  $B_n$  attaining Theorem 12 and put  $\sigma_n = M(B_n) - F(n)$  and  $\delta_n = \delta_w(B_n)$ . Exactly

$$F(n + 1) - F(n) = \sigma_n + (n \bmod 2) + 2\delta_n.$$

Telescoping and  $F(n) = O(n^{3/2})$  show

$$\sum_{n=N}^{2N} (\sigma_n + 2\delta_n) = O(N^{3/2}).$$

Thus, for every  $g(N) \rightarrow \infty$ , all but  $O(N/g(N))$  indices in  $[N, 2N]$  have a Bellman-optimal predecessor with  $\sigma_n \leq g(N)\sqrt{N}$  and  $\delta_n \leq \frac{1}{2}g(N)\sqrt{N}$ . The scalar countermodel below shows that even replacing this density-one statement by uniform  $O(\sqrt{n})$  control would not prove convergence.

### 9.3 Parity gain and the sharp limit of the Gaussian certificate

Put

$$\Delta(A) = \|A^2 - (n - 1)I\|_F^2 = \sum_{i \neq j} (A^2)_{ij}^2.$$

**Theorem 16** (Parity of the Gram defect). *Every order- $n$  signing satisfies*

$$\Delta(A) \geq \begin{cases} n(n - 1), & n \text{ odd}, \\ 2n(n - 2), & n \equiv 0 \pmod{4}, \\ 0, & n \equiv 2 \pmod{4}. \end{cases}$$

*These bounds are attained on infinite Paley-derived subsequences. In particular,*

$$\boxed{F(20) \geq 30}.$$

*Proof.* For odd  $n$ , every off-diagonal entry of  $A^2$  is a sum of the odd number  $n - 2$  of signs and therefore has square at least one.

Now let  $n \equiv 0 \pmod{4}$ , write  $a_{ij} = (-1)^{e_{ij}}$ , and put  $d_i = \sum_{j \neq i} e_{ij} \pmod{2}$ . Reduction modulo four gives

$$\frac{(A^2)_{ij}}{2} \equiv 1 + d_i + d_j \pmod{2}.$$



Thus  $|(A^2)_{ij}| \geq 2$  for every pair with  $d_i = d_j$ . If  $r$  of the  $d_i$  are odd, the handshake lemma makes  $r$  even and

$$\Delta(A) \geq 8 \left[ \binom{r}{2} + \binom{n-r}{2} \right] \geq 2n(n-2).$$

Conference matrices show that the third bound can vanish. Deleting one vertex from a conference matrix makes all off-diagonal Gram entries signs; deleting two makes exactly  $n(n-2)/2$  ordered entries have magnitude two. This proves sharpness on the corresponding Paley orders.

For  $n = 20$ , Theorem 13 and  $\Delta \geq 720$  give

$$M(A) \geq \frac{380}{\pi} \arcsin \frac{1}{\sqrt{19}} + \frac{90}{19\pi 18^{3/2}} > 28.$$

For exact strictness, take  $x = 39/170 < 1/\sqrt{19}$ , use  $1/\sqrt{18} > 70/297$ ,  $\arcsin x > x + x^3/6$ , and  $\pi < 22/7$ . The remaining rational margin is

$$380 \left( x + \frac{x^3}{6} \right) + \frac{5}{19} \frac{70}{297} - 88 = \frac{8798941}{2772405900} > 0.$$

Order-20 energies are even, so  $M(A) \geq 30$ . □

The full parameter family admits an exact Jensen compression. With  $q = n-1$ ,  $\varepsilon = \Delta(A)/(nq^3)$ , and  $0 \leq w < 1$ , put

$$a = \frac{1+w}{2} \sqrt{\varepsilon}, \quad d = \frac{\sqrt{1-w^2}}{\sqrt{q}}.$$

Then

$$M(A) \geq \frac{nq}{2\pi} [\arcsin(a+d) - \arcsin(a-d)].$$

Indeed, after writing  $w = (qs^2 - 1)/(qs^2 + 1)$ , the edge offsets in (25) have mean square  $a^2$ . For fixed  $d$ , the function

$$u \mapsto \arcsin(\sqrt{u} + d) - \arcsin(\sqrt{u} - d)$$

is convex. To see this, if  $f_d(z)$  denotes the arcsine difference, then

$$\frac{d^2}{du^2} f_d(\sqrt{u}) = \frac{zf_d''(z) - f_d'(z)}{4z^3}.$$

The numerator vanishes at zero and has derivative  $zf_d'''(z) \geq 0$ , because  $h'(v)$  is even and increasing in  $|v|$  for  $h(v) = v(1-v^2)^{-3/2}$ . The number  $a$  is the root-mean-square of the edge offsets, so it remains in the correlation domain. Jensen proves the claim; endpoints follow by continuity. Consequently, if  $\varepsilon(A_n) \rightarrow \varepsilon < 1$ ,

$$\liminf_n \frac{M(A_n)}{n^{3/2}} \geq \frac{1}{\pi} \sqrt{\frac{2}{1 + \sqrt{1-\varepsilon}}}.$$

For fixed  $w$ , the limiting factor before optimization is

$$\frac{1}{\pi} \sqrt{\frac{1-w^2}{1-\varepsilon(1+w)^2/4}}.$$

Here the square root multiplies  $1/\pi$ ; optimizing its square gives  $1+w = 2/(1+\sqrt{1-\varepsilon})$  and the preceding formula.

**Proposition 17** (Sharp barrier for the square-covariance family). *If  $B_s(A)$  is the right side of (25) and  $G_n = \min_A \sup_{s \geq 0} B_s(A)$ , then*

$$\lim_{n \rightarrow \infty} \frac{G_n}{n^{3/2}} = \frac{1}{\pi}.$$

*Proof.* Taking  $s = 1/\sqrt{n-1}$  proves the lower bound. For the reverse inequality, take an order- $n$  principal block of a Paley conference matrix of order  $n+r$ , where  $r = o(n)$ . If  $U$  is the deleted block, then  $|(A^2)_{ij}| = |(UU^\top)_{ij}| \leq r$ . Uniformly in  $s$ ,

$$z_{ij,s} \leq \frac{r}{n-1} = o(1), \quad d_s \leq \frac{1}{\sqrt{n-1}}.$$

Writing each arcsine difference as an integral bounds it by  $2d_s(1+o(1))$ , uniformly in  $s$ . Summation gives  $\sup_s B_s(A) \leq (1/\pi + o(1))n^{3/2}$ .  $\square$

#### 9.4 Failure of the weighted union bound and an exact orbit state

**Proposition 18** (Universal entropy floor). *For every pair  $B, D$ ,*

$$L + \sqrt{2nk(\Xi + \log 4)} \geq \sqrt{2nk(n+k) \log 2}.$$

*Proof.* Since  $|\mathcal{R}_{n,k}| = 2^{n+k-2}$  and  $0 \leq w(R) \leq L/2$ ,

$$\Xi \geq (n+k-2) \log 2 - \frac{L^2}{2nk}.$$

Put  $b = \sqrt{2nk(n+k) \log 2}$ . If  $L \leq b$ , the left side is at least  $L + \sqrt{b^2 - L^2} \geq b$ ; if  $L > b$ , the result is immediate.  $\square$

For balanced blocks the floor is  $2\sqrt{\log 2} N^{3/2}$ , whereas the desired right side in (31) is at most  $(\sqrt{2} + o(1))N^{3/2}$ . Thus (31) fails universally by the leading constant  $2\sqrt{\log 2} - \sqrt{2} = 0.250895\dots$ . Even retaining the discarded cross term in the same Hoeffding sum cannot help: for  $K \geq L$ ,

$$2 \sum_R \exp\left(-\frac{(K - h(R))^2}{2nk}\right) \geq 2^{n+k-1} \exp\left(-\frac{K^2}{2nk}\right).$$

There is an exact dependence-sensitive replacement. Regard  $G = \mathcal{R}_{n,k}$  as a group under entrywise multiplication. For a cross seed  $C_0$  put

$$c(U) = |\langle C_0, U \rangle|, \quad h(R) = |Q_B(x) + Q_D(y)|.$$

The row-and-column switching orbit  $C_g = C_0 \odot g$  satisfies

$$J_{\text{orb}}(C_0; B, D) = \min_{g \in G} \max_{R \in G} [h(R) + c(gR)].$$

For  $K \geq L$ , let  $C_t = \{U : c(U) > t\}$ . Its bad shifts are exactly

$$\mathcal{B}_K = \bigcup_{R \in G} R C_{K-h(R)}.$$

Hence the orbit succeeds if and only if  $\mathcal{B}_K \neq G$ . This weighted sumset noncoverage problem on  $\mathbb{F}_2^{n+k-2}$  retains the intersections lost by the union bound.

## 9.5 The exact scalar Bellman wall

Let

$$r_n = F(n+1) - F(n) - (n \bmod 2) = \sigma_n + 2\delta_n.$$

The normalized dyadic cost

$$A_N = N^{-3/2} \sum_{n=N}^{2N-1} r_n$$

is an exact diagnostic:

$$\frac{F(n)}{n^{3/2}} \text{ converges} \iff A_N \text{ converges.}$$

If  $A_N \rightarrow \Lambda$ , the limit is  $\Lambda/(2^{3/2} - 1)$ . Indeed, with  $c_N = F(N)/N^{3/2}$  and  $q = 2^{-3/2}$ ,

$$c_{2N} = qc_N + qA_N + O(N^{-1/2}).$$

The forward implication is immediate. Conversely, for arbitrary  $n$  and  $m = \lfloor n/2 \rfloor$ , the same relation and  $F(2m+1) - F(2m) = O(m)$  give  $c_n = q(c_m + \Lambda) + o(1)$ . If  $C = \Lambda/(2^{3/2} - 1)$  and  $e_n = c_n - C$ , boundedness yields  $\limsup |e_n| \leq q \limsup |e_n|$ , hence convergence at all orders.

Uniform magnitude control of  $r_n$  is not enough. Define

$$g(t) = t^{3/2} \left[ \frac{2}{5} + \frac{1}{25} \sin \log \log(t + e^2) \right],$$

use

$$f(1), \dots, f(10) = 0, 1, 3, 4, 4, 5, 9, 10, 12, 13,$$

and for  $n \geq 10$  let  $f(n+1) - f(n)$  be the least positive integer congruent to  $n$  modulo two and at least  $g(n+1) - g(n)$ . Since

$$0.52\sqrt{t} \leq g'(t) \leq 0.68\sqrt{t} \quad (t \geq 10),$$

the resulting Bellman residual is nonnegative, even, and at most  $3\sqrt{n}$ . Moreover  $f(n) = 13 + g(n) - g(10) + O(n)$ , so its normalized liminf and limsup are 0.36 and 0.44: the phase  $\log \log(n + e^2)$  is unbounded and its successive increments tend to zero. For  $n \geq 10$ ,

$$f(n) \geq 13 + g(n) - g(10) \geq 0.36n^{3/2} + 13 - 0.44 \cdot 10^{3/2} > \frac{n^{3/2}}{\pi},$$

and the finite prefix is direct. The increment ceiling gives the trivial upper bound. Positive tail increments, together with the finite prefix, also give both scalar hereditary consequences proved earlier; the complete inequality check is in `RESEARCH.CONTINUATION.md`. Thus no theorem using only those scalar facts can force convergence. A sufficient replacement is

$$\frac{1}{N} \sum_{n \leq N} \frac{r_n}{\sqrt{n}} \rightarrow a,$$

because Abel summation then gives  $F(N)/N^{3/2} \rightarrow 2a/3$ .

## 9.6 Complete order-nine weighted geometry

Exhausting all 12,346 root-normalized order-nine signings gives the new full frontier

$$\mathcal{B}_9 = \{(12, 0)\}.$$

The 55 root records with  $M = 12$  split as 20 with deficit zero and 35 with deficit one; their 15 switching-permutation classes split as four and eleven.

The records `GH0gmo` and `Gxd?Dc` give a stronger collision. Both have  $M = 14$ , projective energy histogram

$$\#\{|Q| = 2, 6, 10, 14\} = (124, 85, 37, 10),$$

and ordered pair-distance enumerator  $(10, 16, 14, 20, 40)$  for their ten exact maximizers. Their profiles  $(\rho_{\text{ext}}, \rho_w, E)$  are, however,  $(4, 4, 15)$  and  $(3, 3, 17)$ . Hence the energy histogram plus the two-point overlap law of exact maximizers still does not determine extension.

Across the full rooted catalogue, ten histogram groups containing 874 records mix deficits. Adding the maximizer pair-distance law leaves three mixed groups containing 112 records. The complete energy-coloured two-point law

$$T_B(e, f, r) = \#\{([x], [y]) : |Q_B(x)| = e, |Q_B(y)| = f, d_{\pm}([x], [y]) = r\}$$

separates all deficits at order nine. This is a finite computation, not an asymptotic sufficiency claim. The verifier asserts the nauty stream count and digest and recomputes every extension both directly and from the weighted radius.

## 9.7 Complete order-ten finite-temperature phases

For order ten, let  $h_j$  count projective spins with  $|Q_A| = 2j + 1$ . Put

$$z = 4 \sinh^2 t, \quad R_0 = 1, \quad R_1 = z + 1, \quad R_{j+1} = (z + 2)R_j - R_{j-1}.$$

Then  $\cosh((2j + 1)t) = \cosh(t)R_j(z)$ , and the augmented partition function is  $4 \cosh(t)P_H(z)$  with  $P_H(z) = \sum_j h_j R_j(z)$ .

The complete set of 274,668 root-normalized records has 6,012 distinct absolute-energy histograms. Every one of their polynomials coefficientwise dominates at least one of

$$\begin{aligned} P_0 &= 512 + 2816z + 4512z^2 + 3680z^3 + 2120z^4 + 792z^5 + 156z^6 + 12z^7, \\ P_1 &= 512 + 2816z + 5280z^2 + 4704z^3 + 2232z^4 + 576z^5 + 76z^6 + 4z^7, \\ P_2 &= 512 + 2816z + 5792z^2 + 5600z^3 + 2520z^4 + 520z^5 + 40z^6. \end{aligned}$$

Their respective nonzero energy histograms are

$$\begin{aligned} (h_0, h_3, h_4, h_7) &= (180, 180, 140, 12), \\ (h_0, \dots, h_7) &= (108, 88, 96, 84, 60, 48, 24, 4), \\ (h_1, h_2, h_5, h_6) &= (200, 192, 80, 40). \end{aligned}$$

The exact differences factor as

$$\begin{aligned} P_0 - P_1 &= 8z^2(z - 2)(z + 1)(z + 3)(z + 4)^2, \\ P_1 - P_2 &= 4z^2(z + 4)^2(z^3 + z^2 - 10z - 8). \end{aligned}$$

The cubic has one positive root  $\zeta = 3.083872359\dots$ . It follows that, for  $t > 0$ , the global minimizing histogram is  $P_0$  until  $t = \operatorname{arsinh}(1/\sqrt{2}) = 0.658478948\dots$ , then  $P_1$  until  $t = \operatorname{arsinh}(\sqrt{\zeta}/2) = 0.792460762\dots$ , and then  $P_2$ . Adjacent phases tie at the thresholds; at  $t = 0$  every signing ties. The maxima of the three phases are 15, 15, 13, respectively, and the first phase is a conference signing. Thus the adversarial optimizer genuinely changes with temperature. This is an exact finite catalogue result, not an asymptotic phase-transition theorem.

## 10 Negative replicas and exact parameter transport

For uniform disorder  $A$  of order  $n$ , define

$$f_{n,t}(A) = \frac{Z_A(t)}{2^{n+1}(\cosh t)^{m_n}}, \quad \mathcal{G}_n(q, t) = \log \mathbb{E}_A f_{n,t}(A)^{-q}.$$

The denominator is the annealed mean of  $Z_A(t)$ .

**Theorem 19** (Negative-replica supermultiplicativity). *For every  $n, k \geq 1$ ,  $q > 0$ , and  $t \in \mathbb{R}$ ,*

$$\mathcal{G}_{n+k}(q, t) \geq \mathcal{G}_n(q, t) + \mathcal{G}_k(q, t). \quad (\text{NR})$$

*Proof.* Fix internal signings  $A, B$  and let  $C$  be a uniform  $n \times k$  cross signing. Write

$$U_A^\sigma = \sum_x e^{t\sigma Q_A(x)}, \quad U_B^\sigma = \sum_y e^{t\sigma Q_B(y)}.$$

Independent averaging of the cross edges gives

$$\mathbb{E}_C Z_{A,B,C}(t) = (\cosh t)^{nk} P, \quad qquad P = U_A^+ U_B^+ + U_A^- U_B^-.$$

Replacing  $B$  by  $-B$  gives  $P' = U_A^+ U_B^- + U_A^- U_B^+$ , and  $P + P' = Z_A Z_B$ . Convexity of  $z \mapsto z^{-q}$ , first over  $C$  and then over the two orientations, yields

$$\begin{aligned} & \frac{1}{2} \left( \mathbb{E}_C Z_{A,B,C}^{-q} + \mathbb{E}_C Z_{A,-B,C}^{-q} \right) \\ & \geq \frac{(\cosh t)^{-qnk}}{2} (P^{-q} + P'^{-q}) \\ & \geq 2^q (\cosh t)^{-qnk} (Z_A Z_B)^{-q}. \end{aligned}$$

Uniform disorder is invariant under  $B \mapsto -B$ . After averaging  $A, B$ , the factors  $2^q$  and  $(\cosh t)^{-qnk}$  cancel exactly against the normalizers because  $m_{n+k} = m_n + m_k + nk$ . This proves (NR). At  $t = 0$  there is equality.  $\square$

Put  $u = \tanh t$  and write  $f_{n,u} = f_{n, \operatorname{arctanh} u}$  and  $\mathcal{G}_n(q, u) = \mathcal{G}_n(q, \operatorname{arctanh} u)$ . The signed even-Eulerian expansion above gives

$$f_{n,u} = T_{u/v} f_{n,v} \quad (0 < u < v < 1),$$

where  $T_\rho$  is the Boolean noise operator on disorder space. Sharp reverse hypercontractivity for positive functions gives, when  $Q' > Q > 0$ ,

$$\|T_\rho f\|_{-Q'} \geq \|f\|_{-Q}, \quad \rho \leq \sqrt{\frac{1+Q}{1+Q'}} [8].$$

Since  $\|f\|_{-Q} = \exp[-\mathcal{G}_n(Q, u)/Q]$  in this parametrization, on the exact invariant curve  $(1 + Q')u'^2 = (1 + Q)u^2$  this says

$$\frac{\mathcal{G}_n(Q', u')}{Q'} \leq \frac{\mathcal{G}_n(Q, u)}{Q}. \quad (10)$$

This is precisely the curve produced by splitting an extensive system, but the inequality points in the wrong direction for closing (NR).

Fix  $\beta, \theta > 0$  and, for sufficiently large  $r$ , set

$$t_r = \frac{\beta}{\sqrt{r}}, \quad q_r = \frac{\theta}{\tanh^2 t_r} - 1 = \frac{\theta}{\beta^2} r + \frac{2\theta}{3} - 1 + O(r^{-1}).$$

**Conjecture 20** (Power-saving negative-replica transport). *For every  $0 < \varepsilon < 1/2$  there are  $C, \delta > 0$  such that, whenever  $\varepsilon N \leq n \leq (1 - \varepsilon)N$ ,*

$$0 \leq \frac{q_N}{q_n} \mathcal{G}_n(q_n, t_n) - \mathcal{G}_n(q_N, t_N) \leq CN^{2-\delta}. \quad (\text{PT})$$

Only the first inequality is proved by (10). If the upper estimate held and  $a_n = \mathcal{G}_n(q_n, t_n)/q_n$ , Theorem 19 would give

$$a_{n+k} \geq a_n + a_k - O((n+k)^{1-\delta})$$

on balanced splits.

**Lemma 21** (Balanced almost-superadditivity). *Let  $b_n \geq 0$  with  $b_n = O(n)$ . If, for some  $K, \delta > 0$  and every sufficiently large balanced split  $N = j + k$ ,*

$$b_N \geq b_j + b_k - KN^{1-\delta}, \quad N/3 \leq j, k \leq 2N/3,$$

*then  $b_n/n$  converges.*

*Proof.* Fix a sufficiently large  $m$ . Write  $N = qm + r$ ,  $0 \leq r < m$ . If  $r > 0$ , partition  $N$  into  $q - 1$  leaves of weight  $m$  and one leaf of weight  $m + r$ ; if  $r = 0$ , use  $q$  leaves of weight  $m$ . These weights lie in  $[m, 2m)$ . Recursively splitting the number of leaves as evenly as possible, placing the single heavier leaf on the smaller side when needed, gives a binary tree in which each child has between one third and two thirds of its parent's total weight.

At an internal node of weight  $W$ , charge its error  $KW^{1-\delta}$  to descendant leaves proportionally to their weights. Parent weights grow by a factor at least  $3/2$  along each upward path, so the charge per unit leaf weight is at most

$$K \sum_{h \geq 0} ((3/2)^h m)^{-\delta} = O(m^{-\delta}).$$

The total accumulated error is therefore  $O(Nm^{-\delta})$ . Dropping the nonnegative contribution of the one exceptional leaf gives

$$\liminf_{N \rightarrow \infty} \frac{b_N}{N} \geq \frac{b_m}{m} - O(m^{-\delta}).$$

Taking  $m$  to infinity along a subsequence attaining  $\limsup b_m/m$  proves the claim.  $\square$

Here  $0 \leq a_n \leq m_n \log \cosh t_n = O_\beta(n)$ , so the lemma proves convergence of  $a_n/n$ . The exact soft-minimum error is asymptotically at most  $\beta \log 2/(2\theta)$ . Thus (PT) for unbounded  $\theta$  at every fixed  $\beta$ , and for unbounded  $\beta$ , would settle the original question.

The finite evidence is adverse but not decisive. Exact energy spectra and exact labeled class multiplicities through order nine give

| $n$                                      | 4      | 5      | 6      | 7      | 8      | 9       |
|------------------------------------------|--------|--------|--------|--------|--------|---------|
| $\Delta_n/n^2, (\beta, \theta) = (1, 1)$ | .00492 | .00878 | .01166 | .01387 | .01560 | .01700  |
| $\Delta_n/n^2, (\beta, \theta) = (2, 4)$ | .14850 | .27713 | .32759 | .28609 | .30762 | .31597, |

where  $\Delta_n = (q_{2n}/q_n)\mathcal{G}_n(q_n, t_n) - \mathcal{G}_n(q_{2n}, t_{2n})$ . These values neither disprove (PT) nor show an asymptotic gap, but they do not support saturation.

## 11 A known square-order Paley ceiling theorem

The following classical regular-conference construction corrects an overstated external report. It is not a new result [9, 10].

**Theorem 22** (Regular Paley Boolean eigenvector). *Let  $m$  be an odd prime power. A Paley conference matrix  $C$  of order  $m^2 + 1$  has a Boolean eigenvector of eigenvalue  $m$ , and*

$$M(C) = \frac{m(m^2 + 1)}{2}.$$

*Proof.* Let  $K = \mathbb{F}_{m^2}$ ,  $H = \mathbb{F}_m$ , and let  $\chi$  be the quadratic character of  $K$ . Index  $C$  by  $\{\infty\} \cup K$ , with  $C_{\infty, a} = 1$  and  $C_{a, b} = \chi(a - b)$ . Choose signs  $\sigma : K/H \rightarrow \{-1, 1\}$  with  $(m + 1)/2$  positive cosets, so  $\sum_c \sigma(c) = 1$ , and put  $x_\infty = 1$ ,  $x_a = \sigma(a + H)$ .

Every element of  $H^*$  is a square in  $K$ , so the character sum on the zero additive coset is  $m - 1$ . Multiplication by  $H^*$  is transitive on the nonzero cosets of the one-dimensional quotient  $K/H$  and preserves  $\chi$ . All nonzero-coset sums are therefore equal; the total character sum on  $K$  is zero, so each equals  $-1$ . For  $a$  in coset  $c$ ,

$$(Cx)_a = 1 + (m - 1)\sigma(c) - \sum_{d \neq c} \sigma(d) = m\sigma(c), \quad (Cx)_\infty = m.$$

Thus  $Cx = mx$ . The conference identity  $C^2 = m^2 I$  supplies the matching spectral upper bound.  $\square$

These full Paley matrices have normalized Boolean maximum tending to  $1/2$ . This gives no lower bound on the minimax  $F(m^2 + 1)$ . Already  $m = 3$  gives  $M(C) = 15$  at order ten, while  $F(10) = 13$ . The theorem neither improves the known upper bound nor excludes better signings or principal submatrices.

## 12 Six proof programs and the sharp remaining lemma

The detailed derivations and failed variants are preserved in `RESEARCH_LEDGER.md`. The current programs are logically distinct.

**A. Quenched extensive free energy.** Prove  $\lim_n \Psi_{n, \beta}$  for every fixed  $\beta > 0$ , or at least for an unbounded set of  $\beta$ 's. Equation (4) then forces convergence. Theorem 19 gives an exact outer-minimum-compatible recursion, but Conjecture 20 is the missing parameter comparison and its finite data are adverse. Either proving that power-saving transport for unbounded parameters or proving a leading transport gap would materially decide this branch. Proposition 4 separately shows why independent annealed cross blocks fail. A repair must absorb the cross term in a joint minimax order parameter, as random-disorder spin-glass interpolation absorbs it only after disorder expectation [3, 4].

**B. Near-saturated relaxation.** Find a nonconvex compact relaxation whose optimizers have  $\sum_e (1 - a_e^2) = o(n^2)$  and an asymptotic composition law. Then Proposition 3 transfers its limit to signs. The cube and sphere calculations prove that the two obvious feasible sets are structurally too loose or exactly discrete.

**C. Signed Eulerian polynomial.** Prove a second-order limit for  $\min_y \log P_y(t)$  after the explicit MacWilliams terms are included. Puncturing and ordinary products fail because of the common parity phase, rank-one cross lift, and macroscopic family of mixed 4-cycles. A quenched coset-weight theorem, rather than an average weight enumerator, is required.

**D. Cavity covering profile.** Control the Pareto set  $(M(B), \delta_w(B))$  uniformly in the near-optimal energy window. Theorem 12 makes this state exact for one vertex, while the order-seven computation rules out the smaller state consisting only of exact maximizers. A useful theorem must either stabilize the normalized dyadic Bellman cost or prove switching-orbit noncoverage for the multivertex rank-one group. Proposition 18 rules out the all-state weighted union bound, so the replacement must exploit intersections among bad constraints.

**E. Second-order limit object.** Construct a rate-sensitive topology for  $A/\sqrt{n}$  in which the Boolean maximum is continuous and sign arrays have an asymptotically lossless composition. Theorem 9 and Proposition 10 rule out ordinary graphons and empirical spectra. Spectral edges, eigenvectors, or the whole induced Gaussian process would have to remain visible.

**F. Nonexistence.** To separate subsequences one needs order-specific lower bounds on  $F(n)$ , not merely constructions with different energies. Conference, random, and planted matrices supply upper bounds or deliberately inferior examples. The small values fluctuate but do not certify a subsequence mechanism.

No route presently closes. The ultimate finite-temperature lemma remains the following; Conjecture 20 is one precise sufficient mechanism for it, not a proved or empirically favored theorem.

**Quenched minimax free-energy lemma.** For every fixed  $\beta > 0$ , the limit

$$\lim_{n \rightarrow \infty} \min_A \frac{1}{\beta n} \log \sum_{x,s} \exp \left( \frac{\beta s Q_A(x)}{\sqrt{n}} \right)$$

exists.

Proving this lemma for an unbounded collection of inverse temperatures settles the MathOverflow question by (4). In coding coordinates it asks for cancellation in  $P_y(\tanh(\beta/\sqrt{n}))$ . Any proof that bounds the mixed cross-block contribution independently incurs a leading error; a successful proof must make that contribution part of the optimized variational structure. The next negative-replica calculation should therefore determine whether  $\Delta_n/n^2 \rightarrow 0$  or stays bounded away from zero, rather than merely extend the table of  $F(n)$ .

## Reproducibility and final status

The exact identity checker, exhaustive small-order search, optional Z3 certificate, expected outputs, deterministic seeds, and certificate boundaries are documented in `verification/README.md`. The literature map is in `LITERATURE.md`; the original and second proof-search ledgers are both preserved.



The adversarial elliptope limit, the code dual and MacWilliams formula, the oriented covariance barrier, the graph-limit obstructions, finite non-heredity, the weighted covering-radius Bellman identity, the exact negative-replica recursion, and the values through order twelve materially narrow the search. They do not prove existence or nonexistence of the original limit.

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